

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence
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COURSE CODE: CSA1710
REG NO:192511352

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Reverse Engineering

Code of Conduct:

*I _PREETHI R(192511352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Preethir

Reverse Engineering a Social Media Content Moderation System

A social media platform automatically flags and removes harmful content within seconds of posting. Reverse engineer the system design by identifying how content data (text, images) is processed using RDD transformations, and explain how intelligent agents contribute to real-time moderation decisions and escalation to human reviewers.

Reverse Engineering a Social Media Content Moderation System

1. System Overview

A social media content moderation system automatically examines posts immediately after users upload them. The system processes different types of content such as text, images, captions, comments, and metadata.

The system uses PySpark RDDs to distribute large volumes of content across multiple processing nodes. AI models analyze the content, while intelligent agents collaborate to determine whether the content is safe, harmful, or requires human review.

The overall process is:

User Post → Data Collection → RDD Creation → RDD Transformations → Text/Image Analysis → Intelligent Agents → Moderation Decision → Remove/Allow/Escalate

2. Content Data Collection

When a user publishes content, the system collects the required information.

The input may contain:

- Post ID
- User ID
- Text or caption
- Image
- Timestamp
- Hashtags
- Comments
- Other metadata

The content is sent to the moderation pipeline immediately after submission so that harmful content can be detected within seconds.

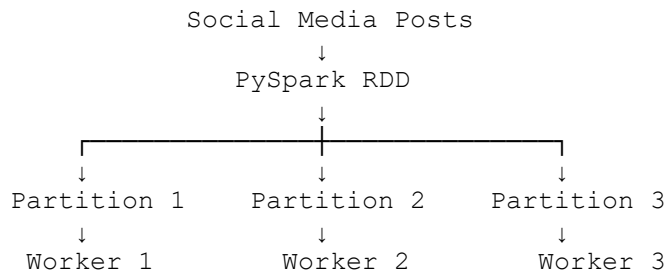
3. RDD Creation

The collected content is converted into a PySpark RDD.

For example:

```
posts = sc.parallelize([
    ("P101", "text content"),
    ("P102", "text content"),
    ("P103", "text content")
])
```

The RDD divides the content into partitions that can be processed in parallel by different worker nodes.



This makes the system scalable when millions of posts are submitted simultaneously.

4. RDD Transformations for Content Processing

RDD transformations are used to prepare and analyze the content.

Map

The `map()` transformation can extract required information from each post.

```
text_data = posts.map(lambda x: (x[0], x[1].lower()))
```

It converts text into a standard format for further processing.

Filter

The `filter()` transformation can remove invalid records or select posts requiring analysis.

```
valid_posts = posts.filter(lambda x: x[1] != "")
```

FlatMap

`flatMap()` can divide text into individual words or tokens.

```
words = posts.flatMap(lambda x: x[1].split())
```

ReduceByKey

`reduceByKey()` can combine or count information by category.

```
word_count = words.map(lambda x: (x, 1)) \
    .reduceByKey(lambda a, b: a + b)
```

These transformations allow the moderation system to process large amounts of content in parallel.

5. Text Processing

Text content is processed through several stages:

Text → Cleaning → Tokenization → Feature Extraction → AI Classification → Moderation Score

The system can examine:

- Offensive language
- Harassment patterns
- Threatening language
- Spam patterns
- Suspicious phrases
- Context and surrounding words

The extracted features are passed to an AI classification model that produces a risk or confidence score.

6. Image Processing

Images require a separate processing pipeline.

Image → Preprocessing → Feature Extraction → Computer Vision Model → Classification → Moderation Score

The image-processing component can identify potentially harmful visual content using computer vision and image classification techniques.

For example, an image can be converted into numerical features and passed to a trained AI model.

The results from image analysis are then combined with the text-analysis results.

7. Intelligent Agent Collaboration

The system uses multiple intelligent agents instead of depending on one component.

Text Analysis Agent

Analyzes captions, posts, and comments to identify potentially harmful textual patterns.

Image Analysis Agent

Analyzes uploaded images using computer vision techniques.

Spam Detection Agent

Identifies repetitive posts, abnormal posting behavior, or automated spam patterns.

Context Analysis Agent

Examines additional information such as hashtags, conversation context, and posting behavior.

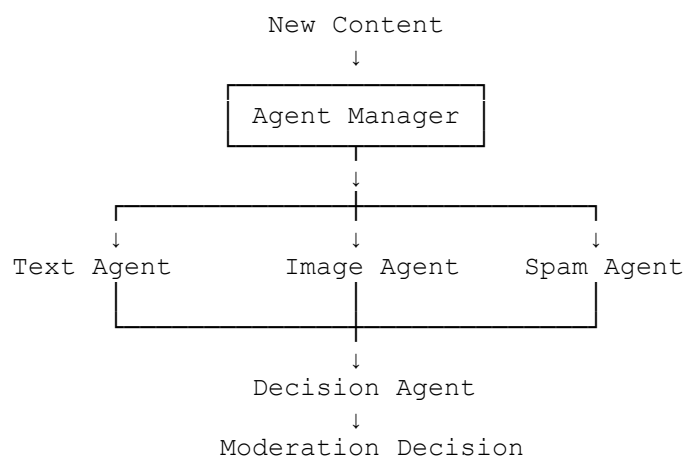
Decision Agent

Combines the results received from the other agents and determines the overall risk.

Escalation Agent

Sends uncertain or sensitive cases to human moderators.

The collaboration can be represented as:



8. Real-Time Moderation Decision

After the agents analyze the content, the Decision Agent combines their outputs.

For example:

Text Risk	= 0.20
Image Risk	= 0.10
Spam Risk	= 0.05

Overall Risk	= LOW

Another post may produce:

Text Risk = 0.90
Image Risk = 0.85
Spam Risk = 0.10

Overall Risk = HIGH

The final decision can be divided into three levels:

Risk Level	System Action
Low	Allow the content
Medium / Uncertain	Send for human review
High	Restrict or remove according to policy

9. Escalation to Human Reviewers

AI cannot always understand context correctly. Therefore, posts with uncertain predictions are sent to human reviewers.

The escalation process is:

AI Analysis → Confidence Check → Uncertain Result → Human Review → Final Decision

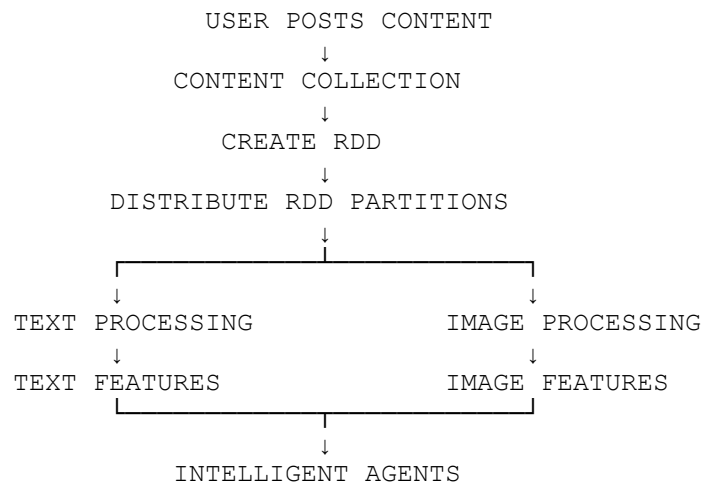
The human moderator can decide whether the content should be:

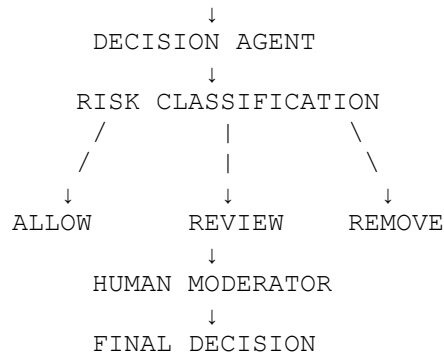
- Allowed
- Restricted
- Removed
- Further investigated

The moderator's decision can also be recorded as feedback for improving the moderation system.

10. Real-Time Processing Architecture

The complete system can be represented as:

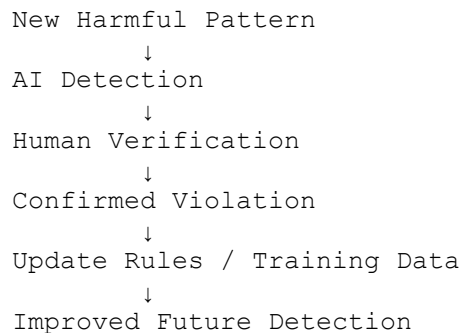




11. Continuous Improvement

The system can continuously improve using feedback from human reviewers.

When a new harmful pattern is confirmed, the pattern can be added to the detection rules or used as new training data.



This enables the moderation system to adapt to newly emerging harmful patterns.

12. Conclusion

The reverse-engineered system uses PySpark RDD transformations to distribute and process large volumes of social media content, while AI models analyze text and images for potentially harmful patterns. Intelligent agents specialize in text analysis, image analysis, spam detection, context analysis, and decision-making.

The Decision Agent combines the results and performs real-time moderation. High-confidence violations can be automatically restricted or removed according to platform policy, while uncertain cases are escalated to human reviewers. Feedback from human decisions can then be used to improve future detection.

Thus, the system provides a scalable, real-time, AI-assisted, and human-in-the-loop content moderation framework.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation